Wattox MPCS-1725-S

P-Q Curve

Prepared By		**Date**	
Reviewed By		**Date**	
Approved By		**Date**	

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Revision History

Date	Version	Description		Prepared By
		Original	Update	
Feb 8, 2025	V1.0	New	Initial version	

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1 Basic Function

In the field of power systems and energy storage, the P-Q capability of a power conversion system (PCS) refers to its ability to control active power (P) and reactive power (Q), a core performance indicator for flexibly regulating the grid or load.

2 Regulatory Requirements

1) Requirements defined in Section 8.1.1.1 "Power Output Range" in GB/T 34133-2023 for power conversion system of electrochemical energy storage system
2) Regulatory requirements of P-Q capability

At the rated voltage and frequency of the AC port, the active and reactive power output ranges of the PCS shall be dynamically adjustable in all the four quadrants within the solid-line area, preferably within the dashed-line area, shown in Figure 2-1.

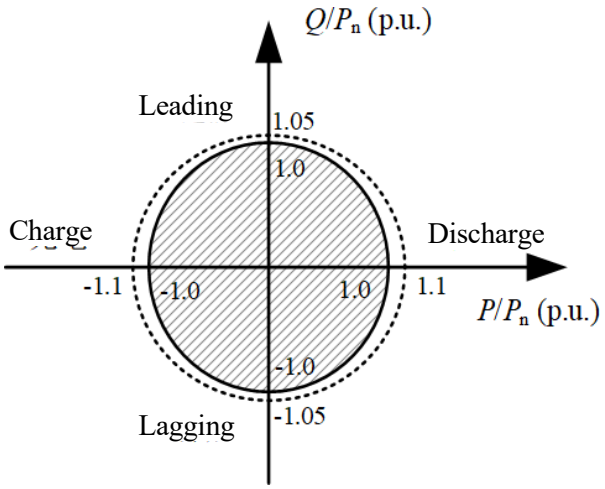


Figure 2-1 PCS power output range

Note: P_n is the rated active power of the PCS.

3 P-Q Power Settings in the Wattox MPCS-1725-S

3.1 Setting Process

Set relevant parameters according to these steps:

1) Set FE-01 (PCS operating mode) to 0 (on-grid mode).
2) Set FE-02 (PCS on-grid control mode) to 0 (P-Q mode).
3) Set FE-06 (preferred power mode setting) to 1 (active power preferred)
4) Set FE-25 (active power setting) to 1 (+: Discharge; –: Charge).
5) Set FE-26 (reactive power setting) to 1(+: Inductive; –: Capacitive).

3.2 Communication Interface Setting

Power of the Wattox MPCS-1725-S can be set with multiple methods: (1) Wattox IPS background; (2) HMI configured for the system; (3) EMS communication ports (RS485, 61850, and MODBUS-TCP).

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